Lecture 12

Stabilizer codes

Objectives. *Lecture 11 gave the exact condition for correcting errors, a matrix of numbers for every pair of errors. This lecture asks:* how do commuting Paulis define a code, and which errors does it correct? *A stabilizer code is the space fixed by commuting Pauli operators; with r independent generators on n qubits it has dimension 2^{n-r} (Theorem 12.3). Its syndromes are bits, read off from commutation, and a set of Pauli errors is correctable if every $E_j^\dagger E_k$ either has a nonzero syndrome or acts as a scalar on the code (Theorem 12.6, the main result). Steane's code is the worked example:* $[[7,1,3]]$.

By the end of this lecture you will be able to:

- Multiply Pauli operators with their phases, decide whether two commute, and compute the syndrome of a Pauli error.
- Prove that a stabilizer group with r independent generators on n qubits has a code of dimension 2^{n-r}.
- Prove that Pauli errors with no $E_j^\dagger E_k$ in $N(S) \setminus \langle i\mathbb{1}\rangle S$ are correctable, and that a code of distance $2t+1$ corrects every error on t qubits.
- Verify from its six generators that Steane's code is $[[7,1,3]]$.

12.1 The Pauli group

Lectures 10 and 11 used products such as Z_1Z_2, $X_1\cdots X_6$ and $Y_i = iX_iZ_i$. With phases allowed, they form a group.

Definition 12.1 (The Pauli group)**.** The *Pauli group* $\mathcal{G}_n$ consists of the matrices $cP_1\otimes\cdots\otimes P_n$ with $c\in\{\pm1,\pm i\}$ and each $P_q\in\{\mathbb{1},X,Y,Z\}$. The *weight* $\mathrm{wt}(E)$ of such an E is the number of q with $P_q\neq\mathbb{1}$. We write $P_1\cdots P_n$ for the product without its phase: $IIXZ$ is X_3Z_4, of weight 2.

By $XY=iZ$, $YZ=iX$, $ZX=iY$ and $YX=-iZ$ (the Pauli relations $\sigma_a\sigma_b=\delta_{ab}\mathbb{1}+i\varepsilon_{abc}\sigma_c$), the product of two elements of $\mathcal{G}_n$ is in $\mathcal{G}_n$, computed qubit by qubit; the phases $\pm1,\pm i$ are what make it closed. Two of $\mathbb{1},X,Y,Z$ commute if one is $\mathbb{1}$ or they are equal, and anticommute otherwise. So any $E,F\in\mathcal{G}_n$ commute or anticommute:

$$EF=(-1)^mFE,$$

where m counts the qubits on which both factors are not $\mathbb{1}$ and differ. For instance $(X\otimes Z)(Z\otimes X)=(XZ)\otimes(ZX)=(-iY)\otimes(iY)=Y\otimes Y$, and $X\otimes Z$ and $Z\otimes X$ commute, as $m=2$. Each Pauli string $P_1\otimes\cdots\otimes P_n$ is self-adjoint and squares to $\mathbb{1}$, so $(cP)^2=c^2\mathbb{1}$: an element of $\mathcal{G}_n$ squares to $\mathbb{1}$ if its phase is ±1, and then it is self-adjoint, and to $-\mathbb{1}$ if its phase is $\pm i$.

12.2 Stabilizers, code space, dimension

Definition 12.2 (Stabilizer code). A *stabilizer group* is a subgroup S of $\mathcal{G}_n$ whose elements commute and which does not contain $-\mathbb{1}$. Generators $g_1, \dots, g_r$ of S (every element is a product of them) are *independent* if none is a product of the others. The *stabilizer code* $C(S)$ is the set of vectors ψ with $g\psi = \psi$ for every $g \in S$; Π_C is its projector.

A vector fixed by the generators is fixed by their products, so $C(S)$ is the common +1 eigenspace of $g_1, \dots, g_r$. As $-\mathbb{1} \notin S$, no element of S has the phase $\pm i$, since its square would be $-\mathbb{1}$: every $g \in S$ has $g^2 = \mathbb{1}$ and is self-adjoint, and $\pm i\mathbb{1} \notin S$. The eight observables of Shor's code (§10.4) generate a stabilizer group: they commute, and a product of some of them that uses an X-product has an X or a Y somewhere (the two X-products multiply to $X_1X_2X_3X_7X_8X_9$), and one that uses only Z-pairs has a Z (the two pairs of a block multiply to Z_1Z_3, and so on); so it is $\pm\mathbb{1}$ only if it uses none.

Theorem 12.3 (Dimension). *Let $g_1, \dots, g_r$ be independent generators of a stabilizer group $S \subseteq \mathcal{G}_n$. Then*

$$\Pi_C = \textstyle\prod_{j=1}^{r} \frac{1}{2}(\mathbb{1} + g_j),$$

and $C(S)$ has dimension 2^{n-r}.

Proof. Each $\frac{1}{2}(\mathbb{1} + g_j)$ is the projector onto the +1 eigenspace of g_j, as g_j is self-adjoint with $g_j^2 = \mathbb{1}$, and they commute; so their product is the projector onto the common +1 eigenspace, which is $C(S)$. Expanding the product,

$$\Pi_C = 2^{-r} \textstyle\sum_{a \in \{0,1\}^r} g_1^{a_1} \cdots g_r^{a_r}.$$

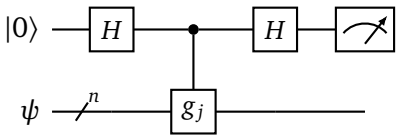
For $a \neq 0$, $g^a = g_1^{a_1} \cdots g_r^{a_r}$ is in S, so it is $\pm P$ for a Pauli string P. If P were $\mathbb{1}$, then g^a would be $-\mathbb{1}$, which S does not contain, or $\mathbb{1}$, which makes each g_j with $a_j = 1$ the product of the others, as $g_j^{-1} = g_j$. So $P \neq \mathbb{1}$, and $\operatorname{Tr} P = \prod_q \operatorname{Tr} P_q = 0$, as $\operatorname{Tr} X = \operatorname{Tr} Y = \operatorname{Tr} Z = 0$. Hence $\dim C(S) = \operatorname{Tr} \Pi_C = 2^{-r} \operatorname{Tr} \mathbb{1} = 2^{n-r}$. ∎

So n qubits and r independent generators encode $k = n - r$ qubits; for Shor's code $n = 9$, $r = 8$ and $k = 1$, the plane of $|0_L\rangle$ and $|1_L\rangle$. Both hypotheses bite. $\langle Z, -Z\rangle$ on one qubit contains $-\mathbb{1}$ and fixes no vector but 0, and the dependent Z_1Z_2, Z_2Z_3, Z_1Z_3 fix the plane of the bit-flip code, of dimension 2, not 2^{3-3}. The proof is Watrous's (2025, pp. 420–421); Nielsen and Chuang prove it otherwise (Proposition 10.5, p. 458).

12.3 Syndromes

An error $E \in \mathcal{G}_n$ commutes or anticommutes with each generator. Its *syndrome* is $s(E) \in \{0,1\}^r$ with $s_j = 0$ if E commutes with g_j and $s_j = 1$ if it anticommutes (Gottesman, (3.9), p. 20; Nielsen and Chuang, p. 466, use the eigenvalues $(-1)^{s_j}$). As in §10.4, for ψ in the code $g_jE\psi = (-1)^{s_j}Eg_j\psi = (-1)^{s_j}E\psi$, so $E\psi$ lies in the range of $\Pi_s = \prod_j \frac{1}{2}(\mathbb{1} + (-1)^{s_j}g_j)$, and the Π_s form the syndrome measurement. Signs multiply, so syndromes add: $s(EF) = s(E) + s(F)$ modulo 2. And $s(E) = 0$ exactly when E commutes with every generator, so with all of S: when E is in the *normaliser* $N(S)$, the set of elements of $\mathcal{G}_n$ that commute with every element of S (as $-\mathbb{1} \notin S$, these are the E with $ESE^\dagger = S$).

Each bit is read with an ancilla:



As $H|0\rangle = |+\rangle$, the controlled g_j and the second H give $|0\rangle \otimes \frac{1}{2}(\mathbb{1} + g_j)\psi + |1\rangle \otimes \frac{1}{2}(\mathbb{1} - g_j)\psi$. By (3.2), the reading gives b with probability $\left\|\frac{1}{2}(\mathbb{1} + (-1)^b g_j)\psi\right\|^2$ and leaves the register in that vector, normalised: the measurement of the eigenvalue $(-1)^b$ of g_j. For g_j a product of Z's, the controlled g_j is a row of controlled Z's, and the circuit can be traded for Lecture 10's CNOTs into the ancilla.

12.4 Distance; correctability

Write $\langle i\mathbb{1}\rangle S$ for the set of cg with $c \in \{\pm 1, \pm i\}$ and $g \in S$.

Lemma 12.4 (Paulis on the code). *Let S have independent generators, $C = C(S)$, and $E \in \mathcal{G}_n$.*

(i) If $E \notin N(S)$, then $\Pi_C E \Pi_C = 0$.

(ii) If $E = cg$ with $g \in S$, then $E\Pi_C = c\Pi_C$.

(iii) If $E \in N(S) \setminus \langle i\mathbb{1}\rangle S$, then E maps C into itself, and is not a multiple of $\mathbb{1}$ on C.

Proof. (i) Some $g \in S$ anticommutes with E, and $g\Pi_C = \Pi_C = \Pi_C g$, as g is self-adjoint and fixes C. So $\Pi_C E \Pi_C = \Pi_C g E \Pi_C = -\Pi_C E g \Pi_C = -\Pi_C E \Pi_C$. (ii) $g\Pi_C = \Pi_C$.

(iii) For $\psi \in C$ and $g \in S$, $gE\psi = Eg\psi = E\psi$, so $E\psi \in C$. Suppose E acts as a scalar on C, so $E\Pi_C = c\Pi_C$. As $E^2 = \pm\mathbb{1}$ and $C \neq 0$, $c^2 = \pm 1$. The operator $F = c^{-1}E$ is in $\mathcal{G}_n$, fixes every vector of C and commutes with S. It is not in S, since $E \notin \langle i\mathbb{1}\rangle S$. Also $F^2 = \pm\mathbb{1}$ fixes C, so $F^2 = \mathbb{1}$.

Adjoin F to the generators. The resulting group S' has elements g and gF with $g \in S$, which commute. It does not contain $-\mathbb{1}$: $gF = -\mathbb{1}$ would give $F = -g$, and then $\psi = F\psi = -g\psi = -\psi$ for $\psi \in C$, contradicting $C \neq 0$. The $r+1$ generators are independent because $F \notin S$.

Compare the dimensions. By Theorem 12.3, the space $C(S')$ has dimension 2^{n-r-1}. But $C(S') = C$, since F fixes C, and C has dimension 2^{n-r}. This contradiction proves that E is not a multiple of $\mathbb{1}$ on C. ∎

The elements of $N(S) \setminus \langle i\mathbb{1}\rangle S$ are the *logical errors*: invisible to the syndrome, and harmful on the code.

Definition 12.5 (Distance). The *distance* d of a stabilizer code is the least weight of an element of $N(S) \setminus \langle i\mathbb{1}\rangle S$, if it has one. A code of n qubits, dimension 2^k and distance d is an $[[n, k, d]]$ code.

Nielsen and Chuang (p. 467) take the least weight in $N(S) - S$, which contains $-\mathbb{1}$, of weight 0; removing the phases ± 1, $\pm i$, as $\langle i\mathbb{1}\rangle S$ does, mends it.

Theorem 12.6 (Correctable Pauli errors). *Let S have independent generators, and let $E_1, \ldots, E_m \in \mathcal{G}_n$. If no $E_j^\dagger E_k$ lies in $N(S) \setminus \langle i\mathbb{1}\rangle S$, then some recovery corrects $E_1, \ldots, E_m$ on $C(S)$.*

Proof. Each $E_j^\dagger E_k$ is in $\mathcal{G}_n$. If it is not in $N(S)$, then $\Pi_C E_j^\dagger E_k \Pi_C = 0$ by Lemma 12.4(i); otherwise it is in $\langle i\mathbb{1}\rangle S$, and $\Pi_C E_j^\dagger E_k \Pi_C = c\Pi_C$ by (ii). So the condition of Theorem 11.2 holds. ∎

Two errors may share a syndrome: $E_j^\dagger E_k \in S$ is allowed, as for Z_1 and Z_2 on Shor's code, where $Z_1 Z_2 \in S$. The code is then degenerate (§11.6), and the recovery need not tell the two apart. What is forbidden is a pair that differs by a logical error. Nielsen and Chuang's Theorem 10.8 (p. 466) asks $E_j^\dagger E_k \notin N(S) - S$, which also excludes pairs such as X_1 and $-X_1$, whose product $-\mathbb{1}$ meets the condition of Theorem 11.2.

Corollary 12.7 (Distance $2t+1$). *If $d \geq 2t+1$, some recovery corrects every element of $\mathcal{G}_n$ of weight at most t, and so undoes every noise whose Kraus operators each act on at most t qubits.*

Proof. For E and F of weight at most t, $E^\dagger F$ has weight at most $2t < d$, so it is not in $N(S) \setminus \langle i\mathbb{1}\rangle S$; Theorem 12.6 applies to these finitely many errors. A matrix on t qubits is a combination of the 4^t products of $\mathbb{1}, X, Y, Z$ on them, so Theorem 11.5 gives the rest. ∎

For general codes Kitaev, Shen and Vyalyi prove the analogue with their distance (Proposition 15.4, p. 160).

Example 12.8 (R as a stabilizer code). Take the repetition code of Theorem 10.4, with

$$\begin{aligned} g_1 &= Z_1 Z_2, \\ g_2 &= Z_2 Z_3, \\ S &= \{\mathbb{1}, Z_1 Z_2, Z_2 Z_3, Z_1 Z_3\}, \\ \Pi_C &= \tfrac{1}{4}(\mathbb{1} + Z_1 Z_2)(\mathbb{1} + Z_2 Z_3). \end{aligned}$$

The two +1 conditions say that all three computational bits agree. Thus C is the span of $|000\rangle$ and $|111\rangle$, of dimension $2^{3-2} = 2$. The two syndrome bits give precisely the old recovery:

Error	Syndrome	Correction
$\mathbb{1}$	00	$\mathbb{1}$
X_1	10	X_1
X_2	11	X_2
X_3	01	X_3

The three cases of Lemma 12.4 now have familiar representatives:

Class of E	R example	Its action on the code
$E \notin N(S)$	X_1	$\Pi_C E \Pi_C = 0$
$E \in \langle i\mathbb{1}\rangle S$	iZ_1Z_2	$E\Pi_C = i\Pi_C$
$E \in N(S) \setminus \langle i\mathbb{1}\rangle S$	Z_1	non-scalar on C

The scalar row retains its phase: $Z_1 Z_2$ fixes C, whereas iZ_1Z_2 multiplies every code vector by i. The logical row has syndrome 00, but changes the relative phase:

$$Z_1(a\,|000\rangle + b\,|111\rangle) = a\,|000\rangle - b\,|111\rangle\,.$$

Hence zero syndrome does not mean harmless. The weight-one operator Z_1 is logical, and every weight-zero Pauli is a scalar. So R has quantum distance $d = 1$: it is an $[[3, 1, 1]]$ code.

This does not contradict correction of the specified error set $\{\mathbb{1}, X_1, X_2, X_3\}$. Correcting a set tests every pairwise product $E_j^\dagger E_k$, not just each error separately. For this set the diagonal products are $\mathbb{1}$. Each off-diagonal product flips one or two bits and has nonzero syndrome, so

$$\Pi_C E_j^\dagger E_k \Pi_C = 0$$

when $j \neq k$. Thus the Knill–Laflamme matrix is $\mathbb{1}$, as in §11.5. Adding Z_1 gives an uncorrectable set: the product $\mathbb{1}^\dagger Z_1 = Z_1$ has $\Pi_C Z_1 \Pi_C$ not a multiple of Π_C. This violates the necessary Knill–Laflamme condition (§11.1; Theorem 11.4). The code corrects single bit flips, but not arbitrary one-qubit errors.

12.5 Steane's code

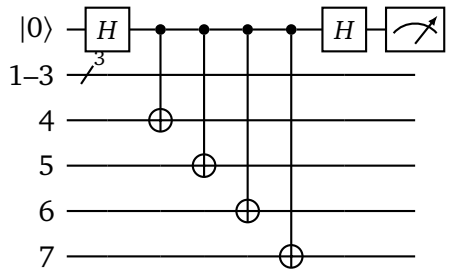
Steane's code has seven qubits and the six generators (Nielsen and Chuang, Fig. 10.6, p. 456, qubit 1 on the left)

$$\begin{aligned} g_1 &= IIIXXXX, \\ g_2 &= IXXIIXX, \\ g_3 &= XIXIXIX, \\ g_4 &= IIIZZZZ, \\ g_5 &= IZZIIZZ, \\ g_6 &= ZIZIZIZ. \end{aligned}$$

Qubit i carries an X in g_1, g_2, g_3, and a Z in g_4, g_5, g_6, according to the binary digits of i, the first generator for the digit 4. An X-type and a Z-type generator share 2 or 4 qubits, so all six commute. Qubits 4, 2 and 1 carry an X in g_1, g_2, g_3 alone and a Z in g_4, g_5, g_6 alone, so a nonempty product has an X, Y or Z on one of them, as $XZ = -iY$. So no nonempty product is $\pm\mathbb{1}$: the generators are independent and $-\mathbb{1} \notin S$. By Theorem 12.3 the code has dimension $2^{7-6} = 2$: $k = 1$.

Syndromes. Z_i anticommutes with the X-type generators with an X on qubit i: its syndrome is i in binary, then 000. X_i gives 000, then i in binary, and $Y_i = iX_iZ_i$ both. So the 21 one-qubit Paulis have distinct, nonzero syndromes.

Each bit is read by §12.3's circuit; for $g_1 = X_4X_5X_6X_7$ the controlled g_1 is four CNOTs:



Distance. A Pauli of weight 1 has a nonzero syndrome. A Pauli of weight 2, A_iB_j with $i \neq j$, has the syndrome $s(A_i) + s(B_j) \neq 0$, as the 21 syndromes are distinct. So $N(S)$ has no element of weight 1 or 2, and $d \geq 3$. And $X_1X_2X_3 = XXXIIII$ is in $N(S) \setminus \langle i\mathbb{1}\rangle S$: it shares qubits 2, 3 with g_5, 1, 3 with g_6 and none with g_4, so it commutes with all six; and the

elements of S whose strings have only I and X are $\mathbb{1}$ and the seven nonempty products of g_1, g_2, g_3, each of weight 4. So $d = 3$: Steane's code is $[[7, 1, 3]]$, and by Corollary 12.7 it corrects every one-qubit error, with seven qubits where Shor's code used nine.[1]

Summary.

- *The Pauli group:* phases $\pm 1, \pm i$; two elements commute or anticommute, by the parity of the qubits where both are not $\mathbb{1}$ and differ; weight.
- *Stabilizer codes:* r independent commuting generators with $-\mathbb{1} \notin S$ fix a code of dimension 2^{n-r} (Theorem 12.3).
- *Syndromes:* bits by anticommutation; they add, and vanish exactly on $N(S)$.
- *The main result:* if no $E_j^\dagger E_k$ lies in $N(S) \setminus \langle i\mathbb{1} \rangle S$, the Pauli errors are correctable (Theorem 12.6); distance $2t + 1$ corrects every error on t qubits (Corollary 12.7).
- *Steane:* $[[7, 1, 3]]$, from its six generators.
- *Next:* Lecture 13, Clifford circuits.

References

Nielsen and Chuang, pp. 454–458 and 465–467. Watrous (2025), pp. 403, 414–428 (little-endian). D. Gottesman, arXiv:quant-ph/9705052 (1997), pp. 15 and 20. Kitaev, Shen and Vyalyi, pp. 160 and 169.

[1] Watrous (2025, pp. 418 and 427) lists the same generators little-endian, each string reversed. Read in our order, his strings give a code in which *XXXIIII* is detected and *IIIIXXX* is the logical error.